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QuantumBlock health records: enhancing healthcare data security with quantum cryptography and blockchain technology

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Abstract

The complexity and cost of contemporary healthcare systems are well-known. Better health record management with advanced technologies needs for mitigating the complex medical systems. Innovation in healthcare delivery with Electronic Health Records (EHR) sharing system is preferred. Throughout the information exchange process, blockchain technology can shield data from hackers. Several traditional methods exist to safeguard electronic medical records. This research aims to propose a novel method in electronic health record security analysis based on quantum cryptography with blockchain models and machine learning techniques. The patient health record data has been analyzed to detect malicious data using a convolutional adversarial transfer encoder neural network. The classification output gives the malicious data, and then the network security is enhanced using quantum cryptography integrated with a federated blockchain algorithm. Experimental analysis is carried out for various EHR datasets regarding training accuracy, mean square error, average precision, data privacy rate, and network security rate. The proposed technique provides a training accuracy of 98%, average precision of 94%, data privacy rate of 93%, Network Security Rate of 95%, and Mean Square Error of 47%. We demonstrate the accuracy and forward security of solutions in EHR scenarios, and our scheme's efficiency compared to other existing schemes is shown by the thorough performance evaluation.

Keywords Electronic health record · Security analysis · Quantum cryptography · Blockchain model · Machine learning techniques

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1 Introduction

With the advent of smart technology, a revolution has begun in several sectors, including the automotive industry, manufacturing, smart cities, and healthcare systems [1]. The healthcare system has evolved from version 1.0 to version 4.0, coinciding with the revolution in several auxiliary businesses. The 1970s–1990s saw the rise of Healthcare 1.0, confined to paper-based generation because digital resources were not yet available. The period known as Healthcare 2.0, which ran from 1991 to 2005, saw the integration of early digital technology with Healthcare 2.0. This generation saw the introduction of patient digital tracking and imaging methods for analysis. Healthcare 3.0 includes electronic health records (EHRs), uploaded to the cloud and readily accessible to physicians via a cloud gateway. The availability of EHRs made communication between the practitioner and the patient more economical and time-efficient. We live in the "Healthcare 4.0" age when healthcare solutions employ blockchain, cloud computing, big data analytics, wearable technology, machine learning, and other technologies [2, 3].

Providing patient-centric healthcare services is the primary objective of Healthcare 4.0. For their healthcare systems, the majority of countries have established a variety of laws and guidelines. Among the essential healthcare products on the market are Healthblocks, AlyaCare, Fronteo, CardioCoin, Kanata, Smart Gate, and Allscripts. These solutions make the current healthcare system more accessible, stable, and long-lasting. Cyberattack risk is increased when multiple technologies are integrated into one platform to serve the healthcare system. There is a potential that eavesdroppers could discover the patient's private information and misuse it, too. Medical equipment was discovered to be the most accessible entry point for an eavesdropper to access EHRs. Nursing staff's frequent access to EHRs increases their vulnerability and provides additional opportunities for an eavesdropper to infiltrate the system [3].

The method administrator is responsible for ensuring security of all linked medical devices. It is, therefore, always a difficult challenge to guarantee security to healthcare method because of the enormous number of devices connected to it. Because they continue to use antiquated technology, certain healthcare systems are more vulnerable to cyberattacks. Security and privacy problems have significantly increased due to expanding use of electronic health records, or EHRs. Patients' privacy is still violated despite the many security and privacy safeguards, frequently due to insufficient privacy rules and dubious information-sharing practices. Millions of records have been made public due to high-profile data breaches in systems like the UK's National Health Service (NHS) and Australia's My Health Record (MHR), which have caused the healthcare sector to suffer significant financial losses [4]. These worries are further heightened by the growing application of machine learning in healthcare for drug development, diagnosis, and precision medicine.

ML models frequently depend on analysing enormous volumes of patient data, including private genetic and clinical data, to reach high accuracy. The widespread use of machine learning (ML) in healthcare industry highlights the need to address the privacy, legal, and ethical issues that arise when AIS, including ML, deep learning, and NLP algorithms, are implemented. Context-sensitive privacy policies are essential for ensuring access controls and privacy settings are carefully tailored to the unique conditions around data [5]. In order to guarantee dependable and trustworthy information sharing in healthcare systems, additional security requirements must be satisfied. These comprise the needs for audits, non-repudiation, secure data transmission, integrity, privacy, authenticity, secrecy, and access control. Current research proposes various security measures to protect healthcare systems' privacy and confidentiality.

Specifically, blockchain technology provides intelligent and complex ways to address security concerns in healthcare systems. According to recent studies, the blockchain helps to securely store sensitive data collected from several sources with better accountability and traceability. Because of concerns about data security or the leakage of patient data during data transfer process, massive EHR distribution has been subject to several restrictions. Furthermore, consumers do not yet have the autonomy to manage their electronic health records (EHRs); instead, EHRs are controlled by hospitals and providers. Blockchain is a great option for EHR and data transmission, providing a secure, decentralised infrastructure for regulated distribution of patient EHRs in healthcare sector.

Blockchain is a decentralised system with sequential links between each block. Healthcare providers, patients, insurance agents, and other stakeholders must work together to administer individual EHR blockchains based on the federal blockchain. A group of parties can offer expedited authentication as well as costly data processing for each contributing member. For instance, the Ethereum blockchain was used in developing the Gem Health Network, designed to minimise the waste of health resources, acquire patient data from various healthcare providers, give people access to data on quick treatment of serious illnesses. Names, patient IDs, and EHRs can all be found in the EHRs recorded in the block. Blockchain technology should be used for patient privacy protection during the EHR distribution process [6].

2 Background study

Modern artificial intelligence techniques are utilised to solve a variety of real-world issues. Numerous classifiers have been developed extensively in this field [6]. Numerous studies deal with keystroke dynamics-based computer-user verification, authorisation, and profiling. Regretfully, several concerns exist regarding the datasets (benchmarks) employed in the study, rendering it exceedingly challenging or unfeasible to juxtapose the findings of disparate investigators. Most typing dynamics approaches focus only on tracking a user's actions as they repeatedly type a set sentence [7]. It is challenging to compare the findings of different studies, even when many are based on lengthy text input, because of variations in data formats, participant numbers, testing durations, and dataset sizes. Most of the time, the studies' data are not openly accessible.

Qu et al. [8] recommended enhancing consensus processes and storing patient medical data on a blockchain to enhance method data security as well as privacy. Author suggested an attribute-based encryption solution for a safe medical record exchange method. Smart contracts ensure integrity as well as traceability of health data. A publication [9] described an attribute-based signature mechanism that uses blockchain to support multiple users in EHR management. They sought to improve system security by employing a decentralised method for enhanced privacy in this attribute-based mathematical formulation.

Alsuqaih et al. [10] suggested a technique design that uses a patient-centric agent in primary model based on a continuous patient monitoring approach. They employed captured simulations to improve privacy as well as security of suggested approach. Al-Saggaf et al. [11] presented a blockchain-based data-sharing protocol, distributed attribute-based signature technique for medical systems, and corresponding algorithms. Pulipeti et al. [12] suggested an EMR access control technique with more granular access inside a method in response to current issues. Liu et al. [13] developed a distributed ledger-based system for electronic medical record security and improved health record interoperability across institutions. Two methods for assessing the system performance of the blockchain architecture were investigated [14]: Configuration endorsement policy and aggressive caching for performance maximisation. The performance indicators of the hyperledger fabric framework were analysed [15]. To optimise performance, Tripathi et al. [16] recommended customising the blockchain framework and design architecture by decreasing input/output and computation. Datta et al. [17] suggested a blockchain-based searchable encryption technique for EHRs. This study developed an indexing method and a two-part evaluation scheme for medical records.

Most research indicates that blockchain is a potential approach to safeguarding confidentiality as well as privacy of personal health data. For example, [18] studied the benefits of blockchain technology in healthcare to provide confidentiality and privacy. Shen et al. [19] created a multiparty consent management system for patient health record (PHR) data exchange. Multiparty authorisation (MPA) technique is used in management to protect patient information when it is shared. Every data-sharing procedure uses the secret key, lowering the overall risks and hazards of sharing. According to experimental results, the suggested MPA maximises accuracy as well as security in PHR data sharing. Suggested approach improves PHR systems' resilience and performance.

Challa et al. [20] suggested a collaborative blockchain-based patient health record (PHR) management as well as sharing plan for healthcare facilities. Hospitals that supply the required patient medical data maintain PHRs. PHR data sharing makes reliable datasets available, which improves the precision of diagnosis and decision-making procedures. Consortium blockchain gives consumers appropriate authorisation and authentication procedures that shield data from hackers. The suggested approach improves PHR systems' dependability and efficiency. Suganthi et al. [21] created the revocable storage and hierarchical attribute-based access control (RS-HABE) method to facilitate data exchange in EHR methods. Proposed RS-HABE method offers a suitable option to guarantee medical data security. Data transmitted during data sharing are secured via the RS-HABE protocol. The suggested plan improves the systems' performance and viability by increasing data sharing accuracy.

Li et al. [22] developed an enhanced blockchain-based Merkle tree electronic medical record (EMR) safe storage system. EMR systems offer users a range of services using the convolutional layer structure. The new plan lowers the complexity and time required to share data and other procedures. The precise content of EMR data is computed using the Merkle tree. Compared to prior programmes, the newly proposed system maximises the security range of electronic medical records in healthcare facilities. Bansal et al. [23] proposed a distributed multilevel electronic medical record (d-EMR) management strategy. This study used blockchain technology to detect security risks that arise from data exchange. The suggested scheme's primary goal is to manage EMR data sharing.

Internet technologies are used in developing parts of the world for online safeguarding and sharing medical certificates with blockchain technology [24]. The research notes data enhancement of integrity, transparency and access within the healthcare sectors. It also emphasizes the security and trust issues associated with managing medical certificates based on the blockchain. Ch. [25] described a

novel blood bank operations management communication structure that delivers data through XGBoost-based predictive access control and is based as a backend provider in real-time. The system guarantees robust and effective data communication between the parties and expedites access to blood collections. It shows how Artificial Intelligence and Blockchain technology can be integrated to improve communication in the healthcare sector.

Rupa et al. [26] studied ways to improve the security of medical certificates stored in the circulation of blockchain networks and create mechanisms that ensure the protection of the privacy of their owners with the aid of cryptographic approaches to defend against leaks. The authors develop a blockchain-based approach for instant access to medical records without revealing patient identity. The work illustrates how blockchain can solve important issues regarding the privacy of medical records. Kusuma et al. [27] explored the use of Ethereum Blockchain Technology to secure and register land records. It brings forth the use of smart contracts in land registration, which is aimed at making the records unchangeable. It describes the usefulness of blockchain technology in enhancing transparency and decreasing malpractices in real estate management. The research provides an ingenious approach to the perennial problems of land record management.

More privacy-preserving techniques for storing and sharing sensitive medical records have been introduced, now using blockchain [28]. The solution developed is based on advanced cryptographic techniques in a decentralized system that works to protect data from unauthorized access. This research further highlights the importance of blockchain technology in enhancing privacy in the healthcare sector.

2.1 Research gap

Due to concerns about data security or the potential for patient personal information to leak during the data transfer process, massive EHR dissemination has previously been subject to a number of restrictions. Furthermore, patients do not currently have the autonomy to maintain their own electronic health records (EHRs); instead, EHRs are under the control of hospitals and providers. To address this issue, this paper suggests a blockchain security framework for effectively and reliably storing and transmitting electronic health records (EHRs), or BSF-EHR. Blockchain technology and the EHR blockchain have already produced standards for identity management and data recording that are already setup. Additionally, the system ensures accountability through transparent information flow by storing the audited proof of all transactions in an immutable distributed ledger. In order to reduce medical errors and safeguard patient privacy, patients can save medical information and discovery data from clinicians in their electronic health records (EHRs). Blockchain is a great option for EHR and data transmission, providing a safe, decentralized infrastructure for the regulated distribution of patient EHRs in the healthcare sector. Blockchain is a decentralized system with sequential links between each block.

The proposed method is as follows: Sect. 3 explains the proposed model in EHR-based data security analysis, and malicious data classification has been carried out using a convolutional adversarial transfer encoder neural network is described in Sect. 4, and Sect. 5 explains quantum cryptography integrated with federated blockchain algorithm.

3 Proposed EHR data security analysis model

This work investigates how our suggested framework satisfies the security requirements of physicians, patients, and other stakeholders, and how structure handles safety and privacy issues in the healthcare industry. The results of the simulation demonstrate how well this system secures EHR data. A patient with a long medical history may also have to face the inconvenience of needing to bring much paperwork to doctor's appointments. Healthcare institutions can digitize the data they gather on their patients, but human verification and database entry will always be necessary. This is an untrustworthy approach to handling such sensitive material. Patients must visit many help desks to obtain the required data using the current record-keeping method. Most of these issues are disregarded, and some are even accepted as standard procedures in our healthcare industry. A patient has also been instructed to reschedule without obtaining a consultation because they forgot to bring the required documents to the appointment, or doctors have administered the incorrect prescription based on an unsubstantiated report.

Blockchain data structures are used by blockchain technology to store and validate data. To ensure that only that person can access the information, it is first encrypted when received from the end user. Each block has key-value pairs containing data; each block is correctly linked to one before and after. The system gets alerted when there is tampering with one of the blockchain's data blocks. The system gets alerted if blocks split up.

All information stored on the blockchain will be completely accessible to all entities and encrypted, meaning no one can examine it until the owner authorizes others to do so. Updating, retrieving, and displaying medical data will be more seamless with a blockchain-based system for storage and retrieval. Therefore, in blockchain architecture,

hospital operations can be categorized as information services. Therefore, one way to conceptualize smart contract design is to compute information service start and completion timings in a blockchain-driven cyber environment that emulates real-world EHR activities. The blockchain architecture is depicted in Fig. 1. It is a group of blocks connected using simple cryptography. The three key concepts of decentralization, transparency, and inflexibility form the foundation of blockchain technologies.

Our system relies on supervised learning, namely classification, which calls for labelled training data. Therefore, we asked two clinical specialists with competence in diabetes to evaluate sample EHRs and classify the samples into case, control, and uncertain categories. We note that our expert review approach is based on hand evaluating each patient's whole record, as is typical for comparable efforts, rather than considering a small number of criteria in our data filtering or baseline expert algorithms.

4 Malicious data detection using convolutional adversarial transfer encoder neural network (CATENN)

Neural networks that mandate the usage of local receptive fields are called convolutional neural networks. The convolutional and pooling layers are two specialist layers used to do this. A CNN typically consists of two phases. Convolutional and pooling layers are initially used to find local input correlations and enable translation invariance. This phase is conceptualized as feature extraction as well as learning from input. The features discovered in the first step are combined in the second stage using fully connected layers to provide an overall forecast of which class input. A graphical depiction of a standard CNN structure for image classification is shown in Fig. 2.

4.1 Convolutional layer

The fundamental component of a CNN is the convolutional layer. By restricting each layer node's scope to a small subset of the previous layer's nodes, a convolutional layer upholds the concept of local receptive fields. As mentioned in the preceding section, weight-sharing—the capacity to identify recurring or translated patterns without introducing more complexity into the network—is a characteristic of local receptive fields. This idea is implemented by a convolutional layer utilizing a filter, a sliding function that finds the same pattern in several input regions. The representation shows a single filter searching through the input for a specific pattern. A convolutional layer usually consists of multiple filters to examine the input for various patterns; the combined output of these filters is called a feature map—the feature map pinpointed the area where each filter discovered its corresponding pattern.

4.2 Pooling layer

The pooling layer accomplishes two things: (i) it reduces the feature map, and (ii) it gives the features contained inside some degree of translation invariance. The feature map's most significant values lie where the filters identify their respective patterns. Strong positive or negative values in the feature map indicate this. The likelihood that the filter has found a pattern increases with the absolute number's size. The pooling layer lowers the feature map's fraction of low values by summing subregions according to their significant values. The most often used pooling feature is max pooling. Like the convolutional layer, it samples subregions; instead of looking for patterns in the input, it takes the most considerable absolute value in that subregion. As a result, the feature map's density of significant values rises, and the locations of the patterns are translated.

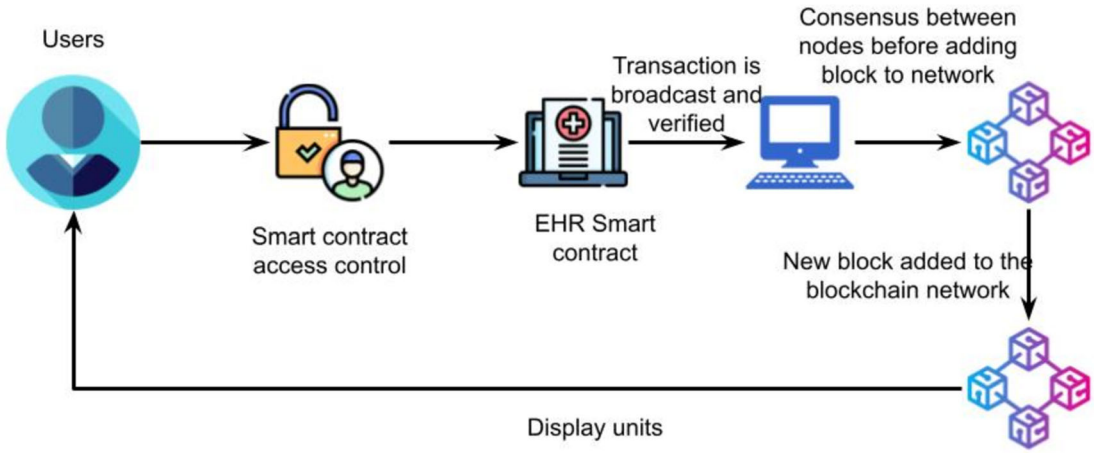

Fig. 1 EHR data security analysis based on a blockchain model

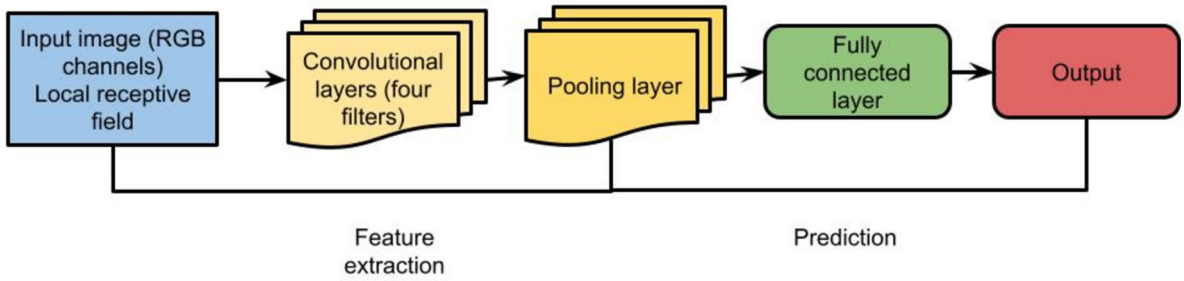

Fig. 2 Convolutional model in data training

4.3 Fully-connected layer

It is present in a fully connected neural network and is also present in a CNN. Every node in this layer is linked to every other layer's node. Global relationships from the previous layer can be extracted from this structure. The different features identified in the convolutional layers are combined by a CNN using fully connected layers to get a final prediction of the entire input. The number of connections in this layer is inversely related to several nodes in the layer above, which is a drawback. Large input sizes can result in significant processing overhead, so convolutional and pooling layers are frequently made to minimize the specifications in the input before using a fully connected layer.

The GAN model's generator and discriminator neural networks compete with one another and play a min–max game to reach a Nash equilibrium point. Using inputs from the input training data distribution, the generator in the unsupervised GAN model aims to generate outputs that closely match this distribution. Conversely, the discriminator's objective is to differentiate between the original (genuine) training data and produced data produced by generator. To train GAN model, we use Wasserstein loss, in which the goal of both discriminators is to reduce greatest Wasserstein-1 distance between generated as well as real data samples by Eq. (1)

$$\min_{\theta_\varepsilon} \max_{\theta_{\mathcal{C}_z}} \mathcal{L}_z(\mathcal{C}_z, \varepsilon) = \mathbf{E}_{z\sim\mathcal{P}_z}[\mathcal{C}_z(z)] - \mathbf{E}_{x\sim\mathcal{P}_x}[\mathcal{C}_z(\varepsilon(x))] \quad (1)$$

where $\mathcal{P}_x$ and $\mathcal{P}_z$ stand for random normal distribution (z $\sim$ $\mathcal{P}_z$ = N(0, 1)) and distributions of actual training data, respectively. The mapping Z → X is implemented using the same process and the objective function that follows is as follows by Eq. (2)

$$\min_{\theta_\mathcal{D}} \max_{\theta_{\mathcal{C}_x}} \mathcal{L}_x(\mathcal{C}_x, \mathcal{D}) = \mathbf{E}_{x\sim\mathcal{P}_z}[\mathcal{C}_x(x)] - \mathbf{E}_{z\sim\mathcal{P}_z}[\mathcal{C}_x(\mathcal{D}(z))] \quad (2)$$

According to both criticisms, weights are controlled and gradients that are not equal to one are penalized through the use of gradient penalty regularisation. Using the autoencoder's characteristics to both encode and decode time-series measurements, AAE model employs two critics to discern between authentic measurements. Let us consider that there are m unlabelled measurements $(X_1, X_2, X_3 \ldots . X_m)$ in the training dataset X. Two criticism, $\mathcal{C}_\mathbf{z}$ and $\mathcal{C}_x$, are used to train encoder as well as decoder, which are parts of GAN generator. While critic Cx is taught to discern between actual input data and decoder's reconstructed outputs, critic $\mathcal{C}_\mathbf{z}$ objective is to assess time-series data's quality in latent space. To rebuild the measurement data with the suggested AAE method, a min–max problem is created by combining all of objective functions found in Eqs. 3.

$$\min_{\theta_\varepsilon,\theta_\mathcal{D}} \max_{\theta_{\mathcal{C}_z},\theta_{\mathcal{C}_x}} \mathcal{L}_{\mathcal{A}\varepsilon}(\theta_\varepsilon, \theta_\mathcal{D}) + \mathcal{L}_z(\mathcal{C}_z, \varepsilon) + \mathcal{L}_x(\mathcal{C}_x, \mathcal{D}) \quad (3)$$

It makes sense to apply knowledge discovered during solution of the natural picture recognition problem to the solution of the medical image classification problem. One transfer learning technique is treating the remaining portion of the network. This layer outputs 1000 class scores for various tasks, similar to the EHR dataset. The collected characteristics are utilised to train a linear classifier. An additional transfer learning technique involves refining classifier on top of network using a new dataset and fine-tuning weights of previously trained networks by extending back-propagation process.

4.4 Examine the following two facts

First, the size of medical dataset is substantially smaller than that of natural dataset; photos in the two datasets come from entirely various areas, meaning their data distributions are different. We use a different type of transfer learning comparable to abovementioned one for modality categorisation. To capture domain-specific features, we modify transfer learning method by retraining only the final full-connected layer(s) of VGGNet-16 and ResNet-50 from scratch while correcting most previous layers to reserve general information. Next, we train CNN-6 from scratch using the medical dataset to extract more domain-specific data. We specifically used the EHR dataset to pre-train

VGGNet-16 and ResNet-50. We add 30 neurons to the final fully connected layer of pre-trained CNNs. Subsequently, we uniformly restore the weights of the final three fully linked VGGNet-16 as well as final layer of ReNet-50 while correcting all other network layers. We train a model using shuffled mini-batches 32 using Admax optimiser and Categorical Cross-Entropy loss function on the EHR dataset. Let X be an n-image medical training dataset. The procedure of training top layer(s) from scratch involves iteratively determining weights w that minimize empirical loss of CNN by Eq. (4)

$$L(w,X) = \frac{1}{n}\sum_{i=1}^{n} l\left(f(x_i,w), y_i'\right)$$
$$l\left(y_i, y_i'\right) = -\sum_{i=1}^{C} y_i \log f(x_i,w)$$
$$where\, y_i \in \{1,\ldots,C\}, f(x_i,w) \in [0,1] \wedge \sum_{i=1}^{c} f(x_i,w) = 1 = 1 \tag{4}$$

When current weights are applied to the mini-batch, gradient of loss L is used to compute updated weights. Admax is utilized to calculate the unique adaptive learning rates for regulating the weight updates' size. In terms of computation, representation, and statistics, the combination of learners might be advantageous. The EL uses a weighted voting-based approach in this study. T basic learners {h_1, h_2, h_3,…, h_T} are included in its integration, where output of hi on example x is $h_i(x)$. Specifically, T = 4 since we choose four TL approaches in this research. Learner hi predicts a label from set of category labels {c_1, c_2,…, c_N}. Finally, definition of the entire EL process is found in Eq. (5)

$$H(x) = c_{\arg j \max \sum_{i=1}^{T} w_i h_i(x)} \tag{5}$$

where wi is weight of hi; typically, $w_i \geq 0$ and $\sum_{i=1}^{T} w_i = 1$. H(x) is final EL result. We assumed that an independent dataset with m labelled individuals would be represented by the values x_1, y_1, x_2, y_2,…, x_m, y_m. This dataset, which is separate from data in offline learning, is referred to in this work as FC patterns from persons with ASD and healthy subjects. A softmax regression model was then trained using the labels and the high level features that had been extracted. Softmax regression method predicts probabilities y = j z_i, j ∈ [1,…, 5] for a 5-class issue given a high-level feature z_i. Softmax regression hypothesis is calculated using the formula below Eq. (6)

$$h(z_i) = \frac{1}{\sum_{j=1}^{\Pi} e^{\theta_j^T z_i}} \begin{bmatrix} e^{\theta_1^T z_i} \\ e^{\theta_2^T z_i} \end{bmatrix} \tag{6}$$

A vector with five probabilities measuring likelihood of input data for every class label is hypothesis's output. There are just two categories in the current work (5 = 2). The model's coefficient vector, denoted by θ in Eq. (7), can be optimized through the reduction of the cross-entropy cost function.

$$E(\theta) = -\frac{1}{m}\sum_{i=1}^{m}\sum_{j=1}^{\Pi}\left[h(z_i)_j \ln y_{ij} + \left(1 - h(z_i)_j\right)\ln\left(1 - y_{ij}\right)\right] \tag{7}$$

where y_{ij} softmax regression model's node j is output for sample i.

5 Data security modeling by quantum cryptography integrated with federated blockchain algorithm

Numerous blockchain techniques result from conventional efforts to increase healthcare system security. However, the problems of inadequate data management, growing costs, and slower processing speed all persist. The suggested method improves the security as well as confidentiality of healthcare data by utilizing an advanced blockchain method. The main objective of this effort is to use cybersecurity methods based on blockchain technology to enable safe data flow in healthcare methods. The improved security architecture is created with the objective in mind of creating reliable data transmission between patients and the healthcare system.

We have researched the many invasions and cyberattacks that seriously impair healthcare systems nowadays. Additionally, their traits and purposes examine the effects of security flaws and threats. Furthermore, a few potential security fixes from these previous studies are investigated. Based on this analysis, it is more beneficial for us to apply blockchain-based security methods to healthcare methods, which provides a comprehensive picture of cyber-security in healthcare applications. One of the newest technologies being used extensively in several fields to ensure data security and confidentiality is blockchain. Its immutability, persistency, privacy, and decentralisation attracted significant attention in the healthcare sector. Healthcare services are now provided online advances in internet technology, but this opens the door to several security risks, including scalability, interoperability, scattered data, and security breaches. Nowadays, healthcare facilities use centralised database systems to store patient health data. Medical professionals strongly favour centralised storage solutions for health data management since dispersed storage systems are typically more costly in time and money.

However, this is also a memory-intensive job because the medical data needs to be encrypted before being stored in cloud systems. Recent evaluations have evaluated the numerous blockchain-integrated healthcare apps built in traditional works.

Additionally, it demonstrates the popularity and application of blockchain technology in healthcare, where the primary focus is on data security, privacy, and authenticity. Additionally, blockchain solutions are used by most current security frameworks to guard patient privacy against unauthorised access. However, the current studies are severely hindered by the system chain's high complexity, storage cost, lengthy processing times, and slow speed. Therefore, the suggested work encourages the implementation of a blockchain architecture for healthcare security that is both lightweight and computationally efficient.

In light of this, the system employs techniques for intelligent key generation, trust evaluation, and optimisation. For the hospital context, the proposed architecture uses a private blockchain, restricting access to data to those who are authorised. It also effectively speeds up the processing of remote monitoring while safeguarding patient data, diagnostic information, prescription information, and other facts. Figure 3 shows the suggested cybersecurity paradigm for healthcare systems.

Compared to traditional blockchain, Table 1 shows that quantum block data structures have a header and a body. The information required for mining is contained in a block's header. There is a hospital records directory in the current block's body. We deal with susceptible medical record data, so we risk losing the data if the qubits enter superposition. Since the Z gate is a unitary gate that operates on a single qubit, it will help connect the produced Qblocks (quantum registers).

More precisely, it retains the value of 0 unchanged while changing the 1 to − 1. A 180-degree revolution about Z axis of the qubit is used, which leads to a shift in the phase of the qubit. The array that comes before it explains how Z-gates work: Eq. (8) uses two bracket vectors to represent the computational bases of 0 and 1 in the case of qubits.

$$|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \tag{8}$$

The computational base of 1 in the aforementioned matrix has been inverted to -1, indicating that nothing will happen when the qubit is at |0>. The qubit avoids the superposition state and does not change its state once it has been measured.

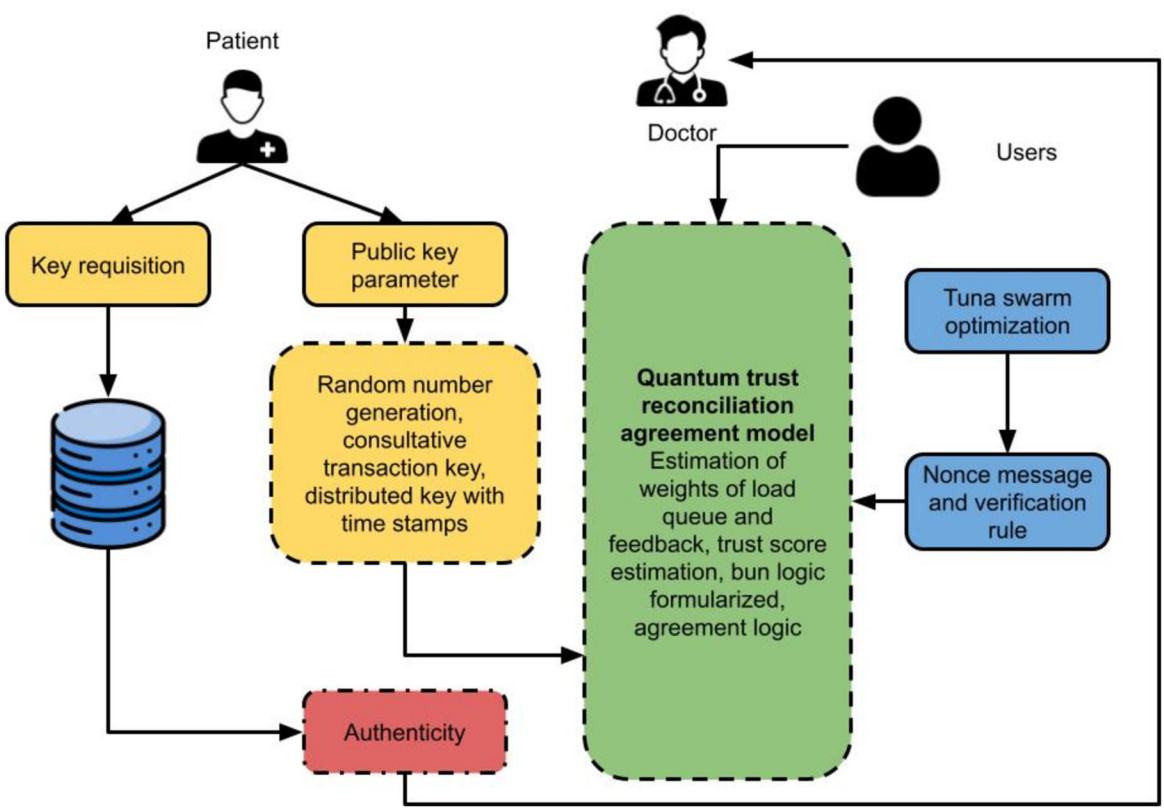

Fig. 3 Proposed quantum blockchain model

Table 1 Quantum Classical Blockchain Data Structures

Blockchain		QMEDCHAIN	
Block header	Block body	Quantum header	Quantum body
Version	List of medical	MQubits	QMedical records
Difficulty target	X	X	X
Merkle root	X	Qhash	X
Hash and timestamp	Records	X	X
Nonce	X	Quantum state	X

A blockchain is a collection of linked blocks connected via cryptography, much like a decentralized database. Every transaction contains a Merkle tree, a hash function, and other elements. In this paper, we secure the credibility of the search process by substituting blockchain for search party in searchable encryption. To guarantee that the blockchain network can rely on a method written in smart contract realizing retrieval of keyword ciphertext, as illustrated in Fig. 3, our paper optimizes and modifies the original blockchain's five-layer architecture and adds a data retrieval function to the application layer.

Patients observe doctors like the conventional EHR procedure. (2) The doctor attends to the patient. (3) A blockchain network consists of nodes, including the EHR system server. It gathers transactions (EHRs) and arranges them into blocks in a mining-like manner. All nodes in the network receive and validate newly produced EHRs following therapy. Subsequently, these transactions are collected by the miner node from memory pool and put together into a block. Every node keeps its memory pool as a "waiting area" for transactions. A transaction waits in memory pool until it is selected by a miner and included in a block after being validated by a node. The blockchain will then include this new block. The miner must first validate information in block before it is added to the chain. This is accomplished by generating a "hash." A hash is a 256-bit value that serves as a block's unique identity for the data. (4) The miner disperses the newly created block to every node in the blockchain network that is accessible (patients and physicians).

5.1 Phase 1

This phase is known as the registration phase, and its goals are to protect patient privacy and provide them with complete control over their data. Patient's true identity is input for this stage, output is creation of the patient's blockchain address, which is utilised as a visit card to prevent patient's identity from being revealed. The system manager generates the private and public keys for the patient and physician in this stage. As a result of this phase, the method manager produces keywords from a critical server.

5.2 Phase 2

Data storage aims to encrypt EHRs with patient identification as well as send them to a private blockchain. Additionally, a smart contract in consortium blockchain will get encrypted index of a doctor's public key, which will be encrypted using the PKES scheme. This stage inputs the patient's address and EHR, stores encrypted EHRs as well as creates a secure encrypted index as outputs.

5.3 Phase 3

Through data sharing, the goal is to give patients authority over a secure blockchain-based EHR sharing method. A third party provides a trapdoor containing a patient's private key and keyword to do this. Outputs send ordered EHRs to a third party, The system manager uses proxy re-encryption (PRE) to renew the keyword set after a comparable epoch and re-encrypt index using patient's private key after every decryption.

Assume a prime finite field Fp and an elliptic curve EC. Assume that P and Q stand in for two random members of group G. Given a line l containing random variables P and Q, we get a tangent line to elliptic curve EC if P = Q and another point R at point where l and curve intersect. Let l be an additional line that runs parallel to y-axis and connects R. When line l intersects curve EC to yield a point R, we define P + Q = R Ec. A scalar multiplication is sometimes defined as tP = (P + P + … + P). For those who are interested, the specifics of the complexity assumptions for computational Diffie-Hellman (CDH) issue and Discrete Log (DL) problem are available here.

In order to create blockchain accounts for patients, physicians, and other data users (third parties), the patients' information had to be registered. The medical service provider, the system manager, the patient who owns the data, and lastly the data consumers (third parties) are the four entities that we use in the system. This stage proceeds as follows. · KeyGen(λ): Returns a pair of public and private keys (Sk,Pk) for a particular patient and doctor as an output and a security parameter (λ) as an input; · PEKS (Pkpi, ω): an input consisting of a chosen collection of

keywords (ω = (ω1,…, ωn) and a patient public key Pkpi. · Trapdoor (Skpi, Q): Takes as inputs patient's public key to create a searchable encryption cω for ω; patient's private key Skpi and keyword query Q = (Ω1,…, Ωn). · KeyRenew (ω,e): It must be performed just once during an epoch.

Using the patient's address, doctor first logs onto Ethereum blockchain. Second, a private blockchain stores the encrypted patient data and the patient's identity-containing EHR. Thirdly, he stores the secure index on the Ethereum blockchain, a consortium-based smart contract, and encrypts it utilizing public-key encryption method—in this case, ECDH algorithm—with doctor's public key. This stage utilises subsequent method: · PEKS (Pk, ω, IDpi): ascertain the location of the EHR in the private blockchain by selecting a keyword set ω = (ω1, ω2,…, ωn) as input and returning it to patient identification IDpi. This is done using the patient public key Pkp; · EncryptEHR (m, Pkp): This function generates ciphertext Cm from the patient's plaintext m and public key Pkp; · Encryptindex(ω,Sk,Pk): This function returns index ciphertext after encrypting keyword set with patient's public key as well as patient's private key.

Initially, Ethereum smart contract receives a request from third party to show the patients' EHRs via trapdoor. Second, the patient's public key from ECC method will be used by EHRs' index to decrypt as well as re-encrypt data. Thirdly, encrypted EHRs index is restored in smart contract, original EHRs are delivered to a third party. · Test(c, ω, Pk, TQ): A public key Pk, a trapdoor TQ, and searchable keyword encryption c are the function inputs. The server outputs "yes" if Q is present in cω and "no" otherwise; · ReKeyGen(ω, sKpi, pKpi): Patient's private key sKpi and the collection of keywords from Ksi are the inputs. This procedure uses the patient's public key, pKpi, to create a secure re-encryption index; · ReEnc(Cm,RKi): Re-encryption secure index (RKi) and ciphertext Cm are the function's inputs, and re-encryption ciphertext Ć m is the output; · Dec(Ć m, Ska): The function's output is plaintext (m), its inputs are patient's private key and re-encryption ciphertext Ć m.

6 Security analysis

6.1 Privacy preservation

Even though every blockchain transaction is visible to the public, the suggested protocol is protected from the sensible IDs of every entity in the system being leaked. We create an authorization token called Apass and leave out the medical doctor's address, as shown in Algorithm 5. As a result, even if a rival obtains the contents, he cannot identify the doctor in real life. As a result, the suggested plan protects patients' and physicians' privacy.

6.2 Authentication

Using the registration credentials, a patient makes an appointment with a doctor under the recommended plan. Every participant receives key pairs Key = {SKUi,PKUi} and establishes an Ethereum account IDUi = {M,Hi, Pi}. As a result, a patient who is not registered cannot even make an appointment with doctor Di via hospital Hi. Consequently, suggested framework ensures that entities are authenticated.

6.3 Authorization

Authorization is achieved by the proposed approach, as patient Pi can only share his EHR data with approved physicians. Pi creates a transaction token on the Ethereum blockchain, Trans = (Rτ ||ST||C1), sends it to physician D2. After then, D2 uses its private key, SKD2, to verify the permission token. A proper verification cannot be accomplished using a non-valid secret key, according to the elliptic curve discrete logarithm problem (ECDLP).

6.4 Confidentiality

Every exchange that takes place between a patient (Pi) and a doctor (Di) is encrypted. Pi delivers authorization token to the doctor Di encrypted as C1 = EncPKD2 Apass. A malevolent user cannot obtain any valuable information from communication, even if he manages to intercept it. As a result, the suggested framework ensures privacy of sensitive patient data.

6.5 Auditability

Suggested plan achieves robust auditability, meaning that in the event that patient EHR records are tampered with, a doctor cannot deny. An opponent attempting to alter the EHR that is shared between the patient and the physician will find it difficult or nearly impossible to do so. To ensure the transaction's immutability, doctor creates a transaction R1 = {T1,Ty1, eh1} and broadcasts it to Ethereum blockchain.

7 Performance analysis

JavaScript implements cryptographic primitives on a Windows 10 PC with an Intel CPU, Intel(R) Core (TM) i7-7500U, 8 GB of RAM. Using Windows 10, a private test blockchain is created utilising the Ganache framework.

Data are programmed into smart contracts and integrated into Ethereum using Solidity programming language. Truffle 5.0.5 is the test framework for smart contracts, and Solc 0.5.16 is the solidity compiler. To connect to blockchain smart contracts as well as assess transaction execution time, Node.js Web3js package is utilised. Solidity still determines how long it will take to publish smart contracts to blockchain. JavaScript can run on the server-side programming platform Nodejs. Table 2 displays the particular setups.

Blockchain: Private Ethereum blockchain network

7.1 Quantum cryptography

Through a Quantum Key Distribution. Use a quantum cryptography library like PyQuil.qskit for simulation. An open quantum-safe library is used for attack detection in the proposed model. We Use the simulated QKD process to generate symmetric keys shared between blockchain nodes. Encrypt the transaction data using quantum-based generated symmetric keys. Deployment is done using a modifying smart contract to accept the encrypted data.

7.2 Neural network training

Using Tensorflow or Pytorch carrying out adversarial deep transfer learning.

Hardware: Cloud environment with combination of GPU for NN training and classical systems required for blockchain operations.

7.3 Dataset description

The patients in the intensive care unit (ICU) must be closely monitored, the ICU is highly data-rich. Researchers are particularly interested in the ICU setting because of the often acute nature of patients' illnesses and the need for prompt management. In particular, many critical care datasets accessible to the public have made study in this field possible. Most of these initiatives are based on MIMIC, a waveform database containing digital transcriptions of over 90 patients' demographic data from paper records. Since all of the clinical data in MIMIC-II was derived from several digital information sources, the sample size and scope of data were much more significant. More recently, in 2015, MIMIC-III was released, greatly expanding on the data from MIMIC-II to include over 40000 patients. The global research community can access several more critical care datasets that are not part of MIMIC initiatives. 208 institutions in the continental United States provide 200859 stays at ICUs and step-down units for eICU Collaborative Research Database (eICU-CRD) v2.0. High-resolution data for nearly 34000 admissions at Bern University Hospital in Switzerland between 2008 and 2016 are available in the HiRID database.712 regularly gathered physiological variables are included in HiRID, with a data entry occurring every two minutes. Children's Hospital at Zhejiang University School of Medicine provides a Paediatric Intensive Care (PIC) database, which includes 13941 ICU admissions and 12881 patients between 2010 and 2018.

Table 3 shows comparative based on various EHR datasets. The datasets analysed are MIMIC-II DATASET, EICU-CRD DATABASE, and HiRID Data. The parameters analysed are training accuracy, average precision, data privacy rate, Network Security Rate, and Mean Square Error. The training and testing have been carried out based on several epochs.

Figure 4 compares MIMIC-II DATASET. Proposed technique attained training accuracy of 86%, average precision of 83%, data privacy rate of 85%, Network Security Rate of 82%, Mean Square Error of 34%; existing RS-HABE attained training accuracy of 77%, average precision of 71%, data privacy rate of 79%, Network Security Rate of 70%, Mean Square Error of 36%, MPA attained training accuracy of 81%, average precision of 75%, data privacy rate of 82%, Network Security Rate of 76%, Mean Square Error of 40%.

Figure 5 compares the eICU-CRD DATABASE dataset. Proposed technique training accuracy of 91%, average precision of 90%, data privacy rate of 93%, Network Security Rate of 89%, Mean Square Error of 40%, existing RS-HABE training accuracy of 77%, average precision of 71%, data privacy rate of 78%, Network Security Rate of 80%, Mean Square Error of 47%, MPA attained training accuracy of 81%, average precision of 85%, data privacy rate of 83%, Network Security Rate of 85%, Mean Square Error of 40%.

Figure 6 (a)–(e) shows a comparison for HiRIDData. The proposed technique attained training accuracy of 98%, average precision of 94%, data privacy rate of 93%, Network Security Rate of 95%, Mean Square Error of 47%;

Table 2 Configurations of Ethereum test blockchain

Component	Description
Operating system	Windows 10
Solidity compiler	Sole 0.5.16
Program language	Solidity and javascript and HTML
CPU	Intel ® core (TM)
Test framework	PTruffle 5.05
Ethereum platform	Ganache v5.4
Interactive platform	Web 3 1.0 0.0 beta 55

Table 3 Comparative for various EHR dataset

Datasets	Techniques	Training accuracy	Average precision	Data privacy rate	Network Security Rate	Mean Square Error
MIMIC-II	RS-HABE	77	71	79	70	36
	MPA	81	75	82	76	40
	CATENN	86	83	85	82	34
eICU-CRD	RS-HABE	82	79	78	80	47
	MPA	84	85	83	85	43
	CATENN	91	90	93	89	40
HiRID	RS-HABE	80	85	82	87	53
	MPA	85	88	86	93	50
	CATENN	98	94	93	95	47

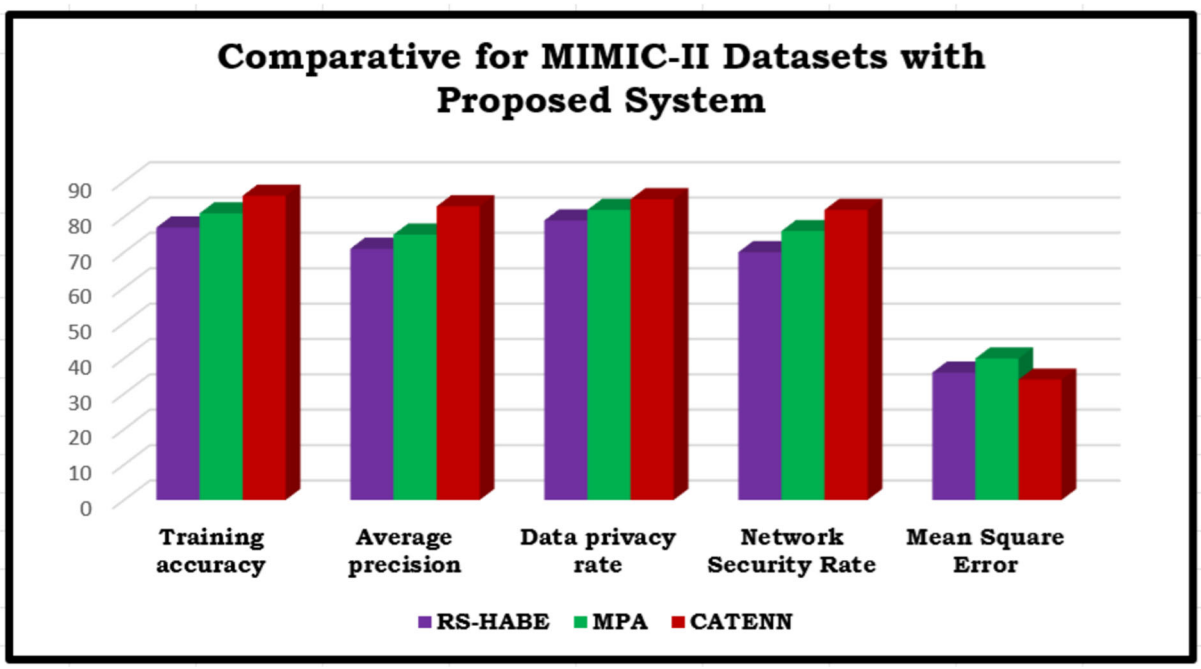

Fig. 4 Comparative for MIMIC-II Dataset dataset in terms of training accuracy, average precision, data privacy rate, Network Security Rate, Mean Square Error

existing RS-HABE attained training accuracy of 80%, average precision of 85%, data privacy rate of 82%, Network Security Rate of 87%, Mean Square Error of 43%, MPA attained training accuracy of 85%, average precision of 88%, data privacy rate of 86%, Network Security Rate of 93%, Mean Square Error of 43%.

We used node.js to construct Hyperledger blockchain network for the PB node in the Docker backdrop. The endorser node and the order node are the two peer nodes that comprise the PB node. Searchable symmetric encryption (SSE)-based techniques Proxy Re-Encryption Scheme (PRES), Proxy Re-Encryption utilising RSA (PRER), TKSE have been put into practice for comparative research. We employ these three cutting-edge methods to compare the performance of the suggested method, changing data size for a fixed set of 20 medical consumers. With publicly accessible EHR research databases, medical data are periodically created from these sources. By altering the data size, (Tables 4 and 5) display the average encryption and average decryption time for every technique.

With a various record size comparable to outcomes, Fig. 7 displays the milliseconds required for the encryption techniques to compute. We measured the time it took to encrypt and upload the record to the blockchain; we changed the record size from 16 to 64 bits to 128 bits to 512 bits. We ran the experiment five times for every record size and then averaged the results. More computing time is

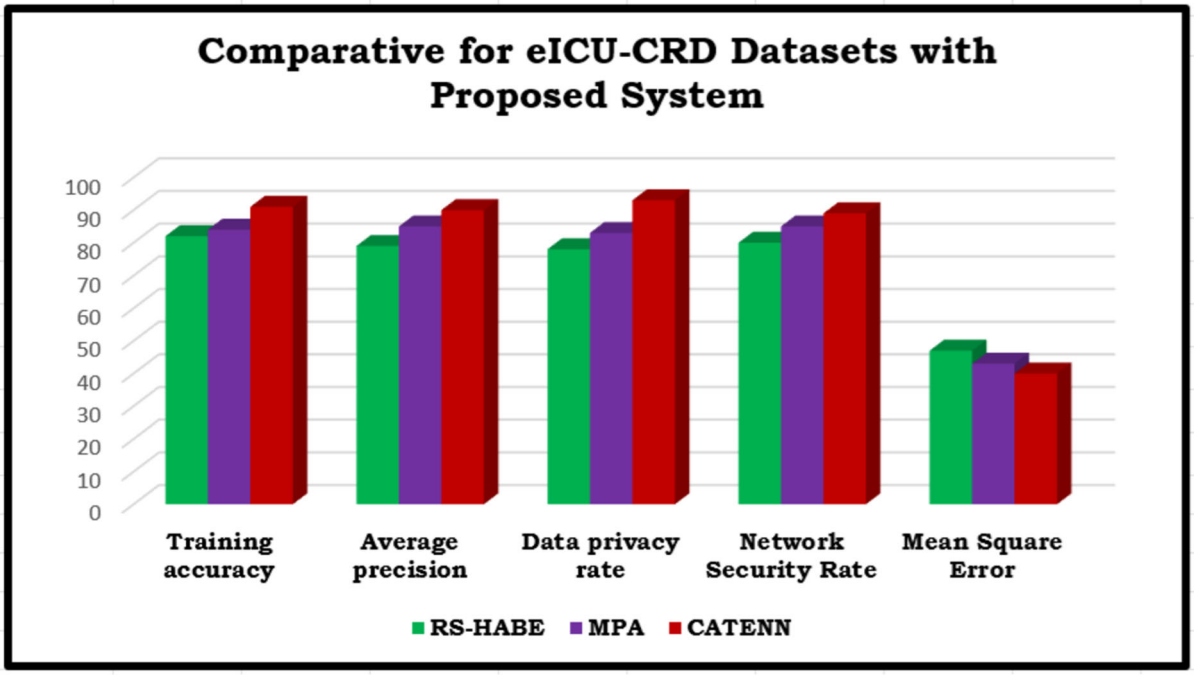

Fig. 5 Comparative for eICU-CRD Dataset dataset in terms of training accuracy, average precision, data privacy rate, Network Security Rate, Mean Square Error

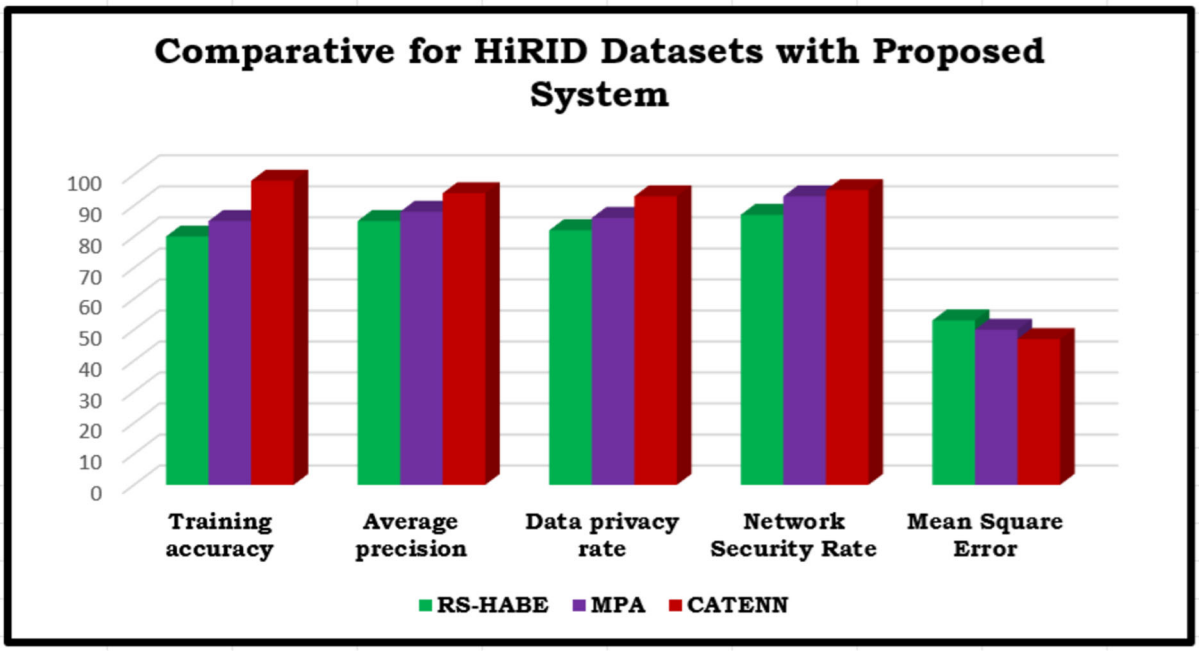

Fig. 6 Comparative for HiRID Dataset in terms of training accuracy, Average precision, Data privacy rate, Network Security Rate, Mean Square Error

required for the more extended record size. Patient and their information determine the size of the record.

We adjust the record size and track how long it takes to encrypt and upload the data to the blockchain. We repeat the experiment five times for every record size, averaging results.

We test different key sizes and record how long it takes to upload the record to the blockchain and encrypt it. We

Table 4 Analysis of average encryption time for different medical data sizes

Data size	TKSE	PRER	PRES
128 KB	645	671	202
512 KB	741	753	232
2 MB	829	845	326
8 MB	910	939	435
32 MB	1626	1678	758
64 MB	4709	4839	3837

Table 5 Analysis of average decryption time for various medical data sizes

Data size	TKSE	PRER	PRES
128 KB	324	352	13
512 KB	741	379	18
2 MB	433	457	48
8 MB	649	682	146
32 MB	1010	1048	647
64 MB	4029	4187	2351

ran the experiment 5 times for every key size before calculating the average.

The computation time of encryption techniques with varying record sizes is shown in milliseconds in Fig. 8. This time, we experimented with several key sizes: 256, 384, and 512 bits. We timed the encryption and blockchain uploading processes for every size. We ran an experiment 5 times for every record size before calculating the average. We can conclude that computation time improves with key size length. Although it needs additional processing time, key size with a maximum length provides reasonable security against attacks.

7.4 The confidence interval of the proposed model

The inclusion of confidence intervals (CI) represents the range of the true metric measurement, allowing assessment regarding the chances and accuracy of results achieved. Calculate the mean value and standard deviation for several measurements taken for each metric of interest, like training accuracy and average precision.

$$CI = \bar{x} \pm z.\frac{\sigma}{\sqrt{n}} \tag{9}$$

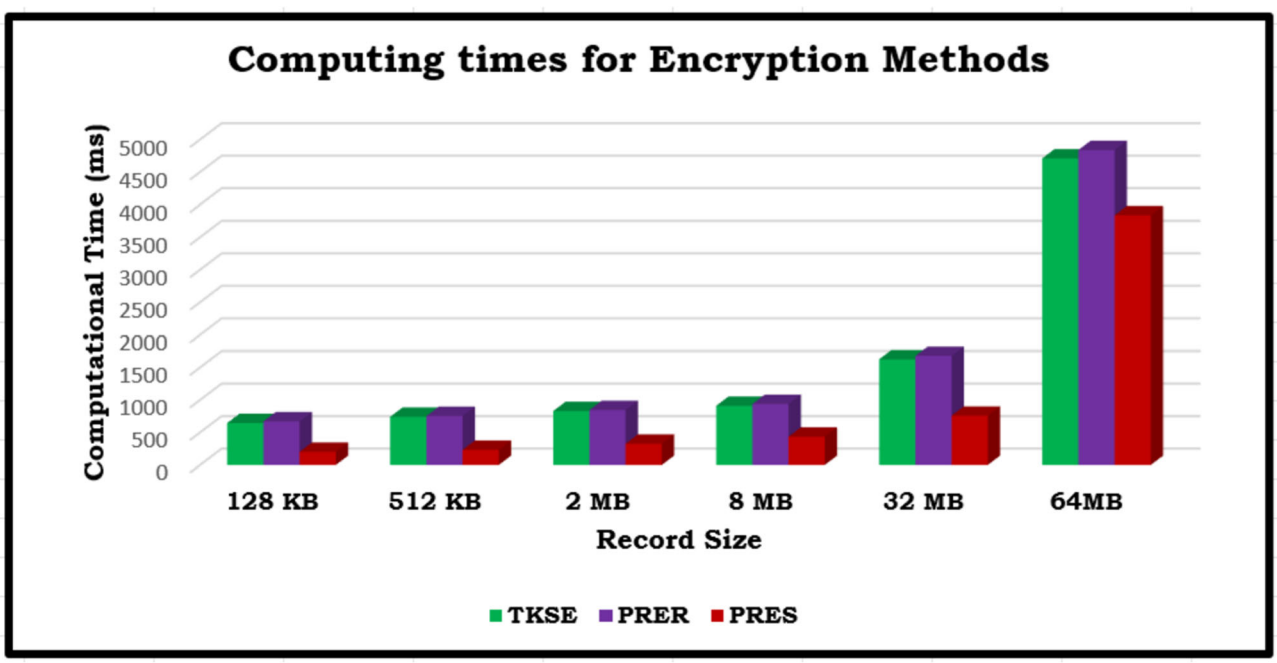

Fig. 7 Computing times for encryption methods with varying record sizes expressed in milliseconds

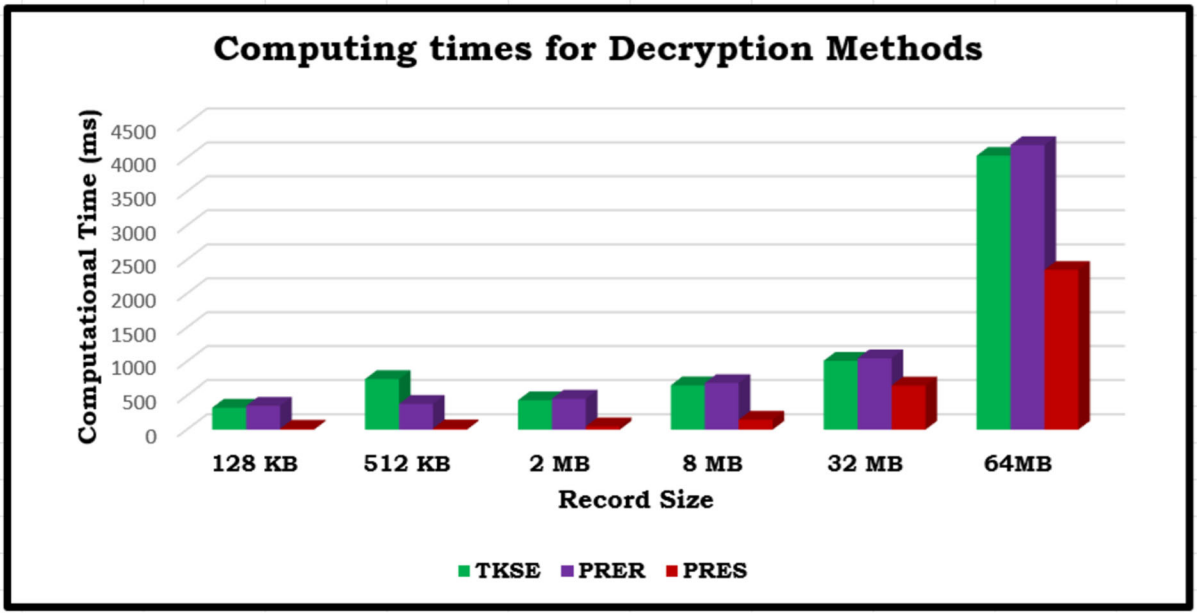

Fig. 8 Computing times for encryption methods with varying key sizes expressed in milliseconds

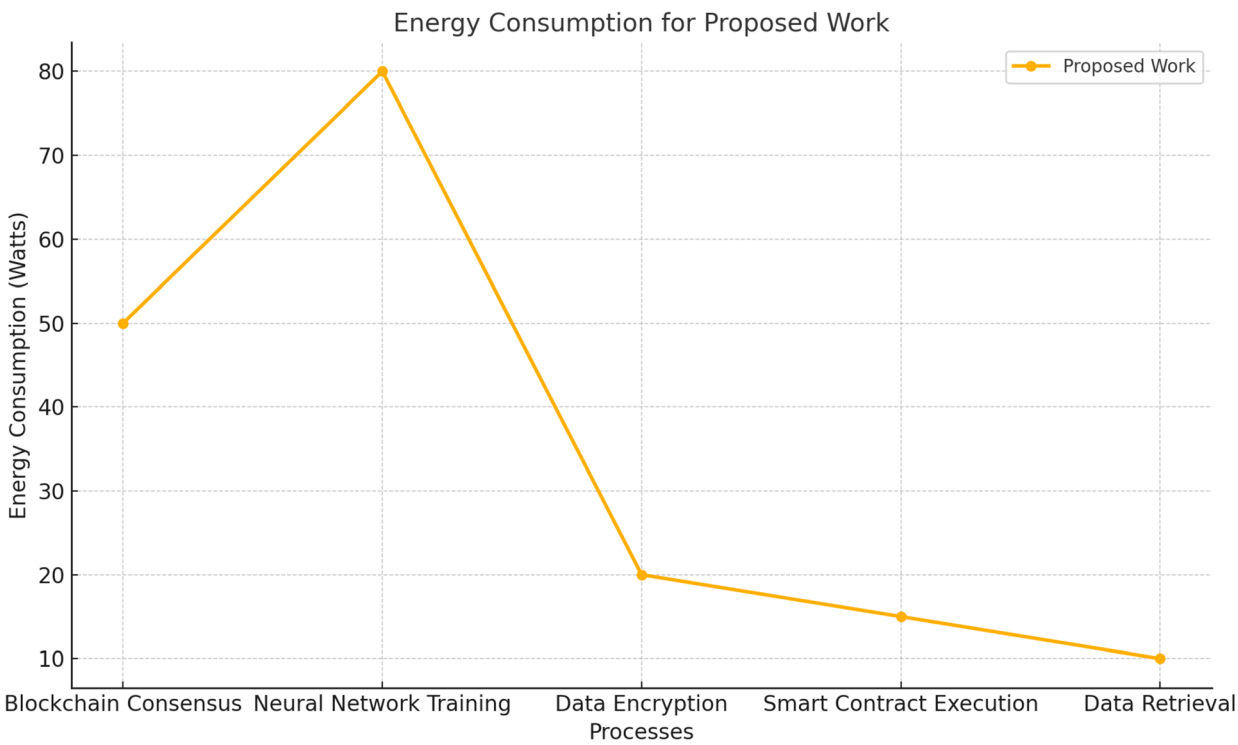

Fig. 9 Energy efficiency in various stages

Figure 9 illustrates the energy consumption of the proposed work across several processes. It allows us to gauge its energy efficiency during blockchain consensus, neural network training, data encryption, smart contract execution, and data retrieval.

8 Conclusion

With the use of blockchain technology, machine learning, and quantum cryptography, this study aims to present new methods for analysing the security of electronic health records. Using a convolutional adversarial transfer encoder

neural network, the patient health record data was examined to identify fraudulent data. The federated blockchain method and quantum cryptography are combined to improve network security when the categorisation output reveals malicious data. Our suggested system enhances security by avoiding outsourcing the search process to a third party and integrating blockchain technology with public-key searchable encryption. Additionally, we guarantee forward security, protecting patient privacy. Patients can safely exchange their private health information with the hospital domain via various safe means. Patients first register with the hospital server using their personal data and medical records. They receive a unique ID and certificate from network management for use in further conversations. The model can fend against quantum attacks, which, if quantum computers are created effectively, might pose a danger to blockchain in the future. It also gives the patients ownership of their data, which is a basic necessity. Our goal of developing a system that can be used by hundreds of thousands of people has been achieved since it employs quantum-safe encryption with little overhead and lightweight, efficient storage. We have proposed and analysed methods. The proposed technique provides a training accuracy of 98%, average precision of 94%, data privacy rate of 93%, Network Security Rate of 95%, and Mean Square Error of 47%. Utilising cryptographic primitives, we ensured that the suggested framework would withstand well-known attacks. The suggested framework's security analysis satisfies all requirements for access control, privacy, and unlinkability. Additionally, the performance evaluation demonstrates the excellent computational efficiency of the suggested framework. This framework is realized and put into practice with the right tools in future studies, and its effectiveness can be assessed, and further enhancements are recommended.

Author contribution P. C: Conceptualization, Methodology, Software, Writing—original draft. L. D: Methodology, Writing— original draft. RKA: Funding acquisition, Data Curation, Validation, Resources, Writing—review & editing B.S.K: Supervision, Visualization A. K: Validation, Writing—review & editing.

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Declarations

Competing interests The authors declare no competing interests.

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